SYSTEM AND METHOD FOR ATTRIBUTED CONTROL AGGREGATION AND INFLUENCE DETECTION

Reinaldo Javier Aguiar Marcano

March 29, 2026

Title: SYSTEM AND METHOD FOR ATTRIBUTED CONTROL AGGREGATION AND INFLUENCE DETECTION

Inventor: Reinaldo Javier Aguiar Marcano, a citizen of the United States, residing at 2302 Britton Ridge Drive, Katy, TX 77494.

Related Documents

The concepts and framework giving rise to the present invention are further described in the whitepaper "Symmetric Governance: Resolving the Recursive Capture Loop through the Decoupling of Time and Capital," (ISBN 978-1-97-148127-2), which is attached hereto and incorporated by reference in its entirety.

Abstract

A computer-implemented method and system for detecting and quantifying covert corporate influence. The system generates a resilient, unique identifier for each entity by creating an "Identity Fingerprint" from its relational and operational data (e.g., past executives, legal representatives, IP portfolio). It then executes an iterative "Influence Smearing" algorithm

that traverses corporate ownership graphs to attribute both direct and indirect control, allowing the total attributed voting power for an entity to exceed 100% to accurately model influence. The system aggregates this influence data across multiple, non-obvious dimensions and records the results in a persistent, immutable storage system to flag entities that exceed regulatory control thresholds in any jurisdiction.

Background of the Invention

Corporate ownership and control are typically tracked using ephemeral identifiers like company names, stock tickers, or registration numbers. Malicious or evasive actors can exploit this system by using complex ownership structures, shell corporations, and frequent name changes to obfuscate their true level of control and influence over publicly traded companies, thereby evading regulatory scrutiny for "Insider" status. Standard accounting and aggregation methods fail to capture these multi-layered, indirect control vectors. There is a need for a system that can pierce the corporate veil and identify true "real interest" and influence, regardless of the obfuscation techniques employed.

Summary of the Invention

The present invention is a computer-implemented system and method for Attributed Control Aggregation and Influence Detection. It provides a novel technological implementation to address the systemic issues of corporate governance described in the related "MARCELO Act" whitepaper.

The method begins by generating a unique, persistent "Identity Fingerprint" for each entity. This fingerprint is derived from a hash of the entity's long-term relational data, such as its key executives, law firms, office locations, and intellectual property. This makes the identifier resilient to superficial changes.

The core of the invention is an iterative "Influence Smearing" algorithm. The system ingests shareholding data and attributes both direct and indirect control. For indirect control, if Entity A controls Entity B, and Entity B controls Entity C, the system attributes Entity C's voting power to both B and A. This process is repeated for N layers of indirection until all non-obvious control vectors have been exposed. The system is designed to measure influence, not just equity, so the total attributed control can exceed the actual number of shares.

The system then aggregates the resulting influence data across dozens of dimensions, including the fingerprints of the law firms, financial institutions, and jurisdictions involved. The results are recorded in an append-only, immutable storage system to ensure data integrity. Finally, the system flags any entity whose aggregated influence exceeds user-defined or regulatory thresholds (e.g., the 10% "Insider" threshold), providing a clear signal of concentrated control for regulators in any jurisdiction.

Statement of License Grant to the United States Government

The inventor, Reinaldo Javier Aguiar Marcano, hereby grants to the United States Federal Government and its military and defense agencies a perpetual, royalty-free, non-exclusive license to practice and use the invention described herein for any and all governmental, defense, and non-commercial purposes. This license explicitly excludes any and all state and local governments.

Brief Description of the Drawings

FIG. 1 is a flowchart illustrating the overall process of the Influence Detection System, from fingerprinting to final flagging.

FIG. 2 is a conceptual diagram illustrating the "Identity Fingerprinting" function.

FIG. 3 is a conceptual diagram illustrating the iterative "Influence Smearing" algorithm for attributing indirect control.

Detailed Description of the Invention

The present invention provides a tangible, computer-implemented system to execute the principles of symmetric governance.

Referring to **FIG. 1**, the overall method is shown. The system begins by ingesting public corporate and shareholding data (101). The first major step is to generate a resilient "Identity Fingerprint" for each entity in the dataset (102). The system then executes the iterative "Influence Smearing" algorithm (103) to attribute both direct and multi-layered

indirect voting control across all entities. Once the iteration stops, the system aggregates the attributed control data across a wide variety of relational dimensions (104). The aggregated results are then recorded into a persistent, immutable storage system (105). Finally, the system compares the aggregated influence levels against regulatory thresholds and flags any entities that exceed them (106).

FIG. 2 illustrates the "Identity Fingerprinting" function. An entity (201) is not identified by its name alone. Instead, the system gathers a set of its long-term relational data, which may include its past CEOs and CFOs (202), its associated law and finance firms (203), its IP portfolio (204), and its physical office locations (205). These data points are fed into a cryptographic hash function (206) to produce a single, unique, and persistent Identity Fingerprint (207) that represents the entity's "real interest."

FIG. 3 illustrates the "Influence Smearing" algorithm. In this example, a primary controlling entity, Vanguard (301), has direct control over Meta (302). Meta, in turn, has direct control over Google (303). In Step 1, the system attributes Google's voting power to Meta. In Step 2 (the first iteration), because Vanguard controls Meta, the system "smears" the influence and also attributes Google's voting power to Vanguard. A counter for Google's shares is incremented to prevent re-counting in subsequent iterations. This iterative process continues for N layers, uncovering hidden chains of influence.

Claims

What is claimed is:

1. A computer-implemented method for detecting and aggregating corporate influence, comprising the steps of:
    - Generating, by a processor, a unique and persistent identifier for each of a plurality of entities by creating an Identity Fingerprint from each entity's relational and operational data;
    - Storing said Identity Fingerprints in a database;
    - Executing, by a processor, an iterative algorithm to attribute voting share control, wherein the algorithm identifies both direct share ownership and indirect control through N layers of intermediate entities;

- Aggregating the attributed voting share control for each unique Identity Fingerprint across one or more dimensions;
- Recording the aggregated control data in a persistent, immutable storage medium; and
- Flagging any entity whose aggregated control data exceeds a predetermined threshold.

2. The method of claim 1, wherein the Identity Fingerprint is generated by applying a cryptographic hash function to a set of data comprising at least two of: a list of past executives, a list of legal and financial representatives, a list of office locations, a list of trademarked products, or a list of patents.

3. The method of claim 1, wherein the iterative algorithm attributes the voting power of a controlled entity to both the direct controller and any entity that indirectly controls the direct controller.

4. The method of claim 3, wherein the algorithm maintains a counter for each block of attributed shares to prevent duplicate counting of an influence vector across iterations.

5. The method of claim 1, wherein the aggregation dimensions are selected from a group comprising: the Identity Fingerprint of the controlled entity, the Identity Fingerprints of the controlled entity's legal and financial representatives, the jurisdiction of the controlled entity, or the year of share acquisition.

6. The method of claim 1, wherein the predetermined threshold is based on a regulatory definition of "Insider" status.

7. A system for detecting and aggregating corporate influence, comprising:
   - A data ingestion module for receiving corporate and shareholding data;
   - A fingerprinting module configured to generate a unique and persistent Identity Fingerprint for each entity from its relational and operational data;
   - An iterative processing engine configured to attribute both direct and indirect voting control across a graph of entities;
   - An aggregation module for summing attributed control across a plurality of dimensions;
   - An immutable storage database for persistently recording the results; and

- A reporting module for flagging entities that either: i) exceed a control threshold (controlling entities), or ii) are being controlled, directly or indirectly, by other entities with an aggregated level of control exceeding a control threshold.

Drawings

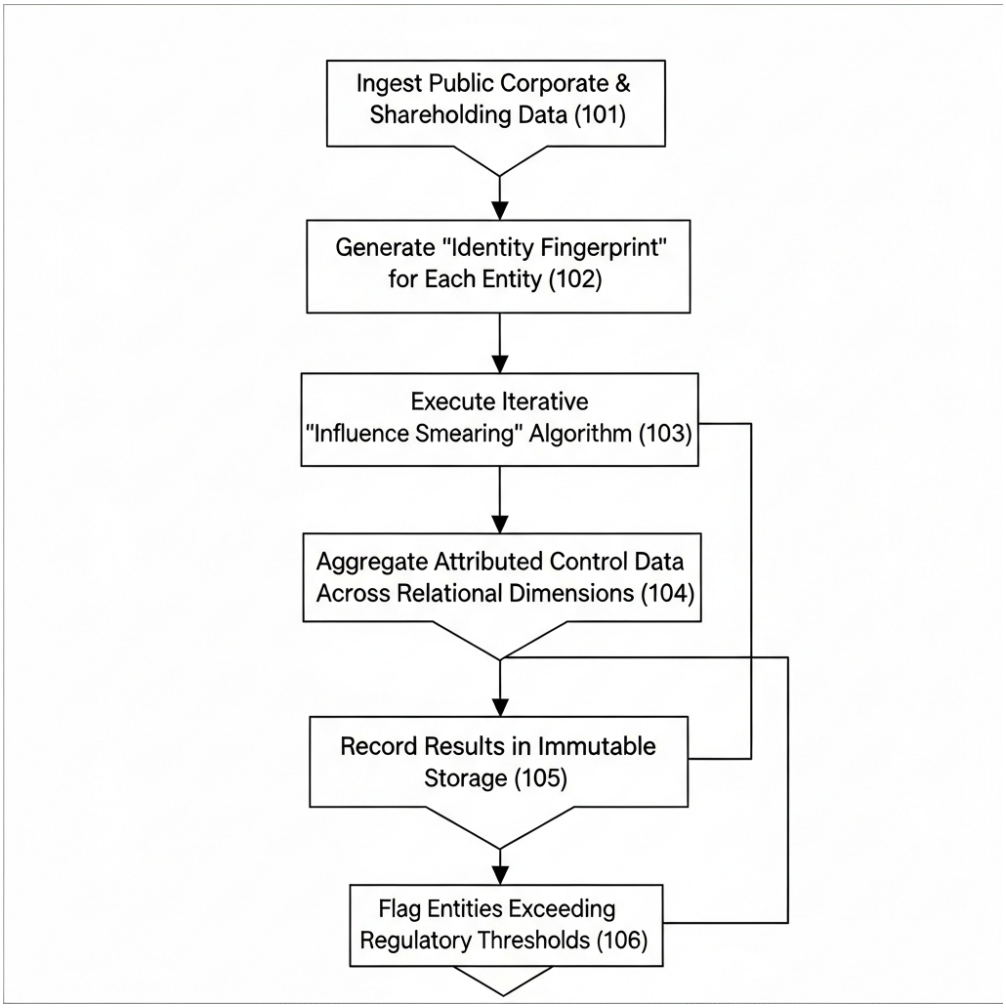


Figure 1: Flowchart illustrating the overall process of the Influence Detection System, from fingerprinting to final flagging.

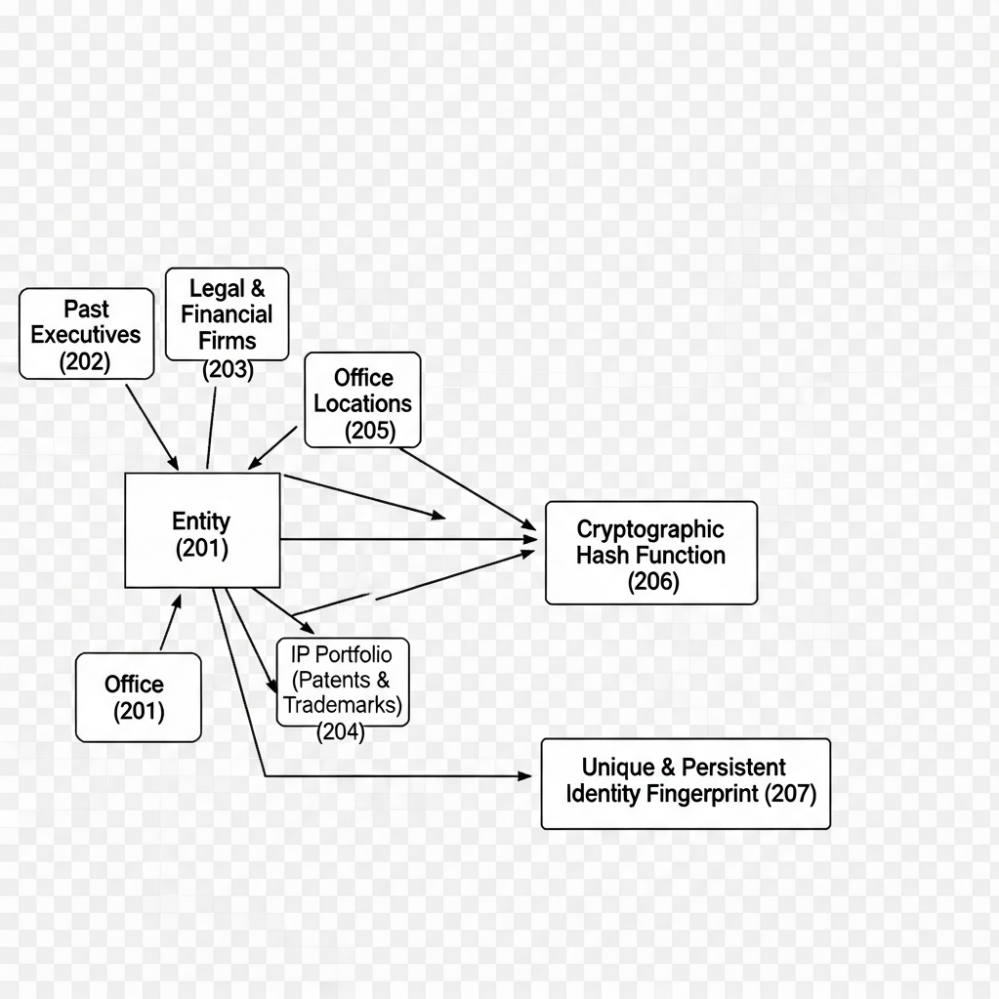


Figure 2: Conceptual diagram illustrating the "Identity Fingerprinting" function.

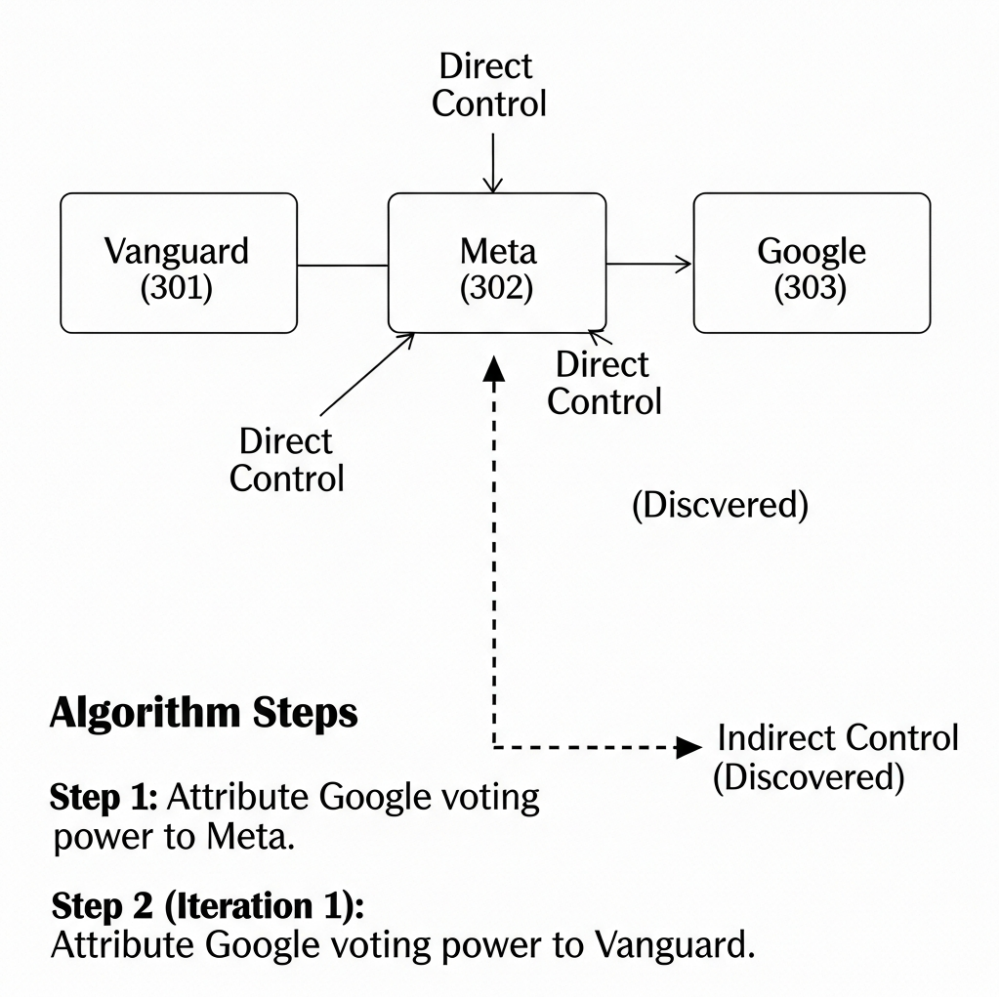


Figure 3: Conceptual diagram illustrating the iterative "Influence Smearing" algorithm for attributing indirect control.

Appendix A - Whitepaper: Aguiar Marcano, Reinaldo J., 2026. "Symmetric Governance: Resolving the Recursive Capture Loop through the Decoupling of Time and Capital." ISBN 978-1-97-148127-2.

width=!,height=!,pages=-